

Nirbhay Singh

Principal FPGA & AESA Radar Systems Engineer | Digital System Architecture | Real-Time Signal Processing

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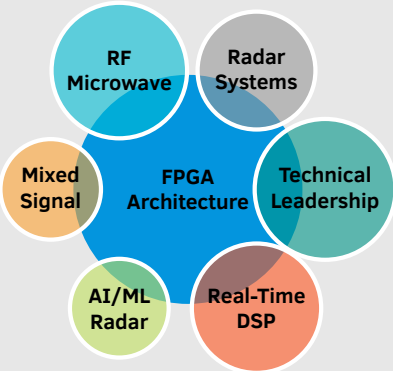
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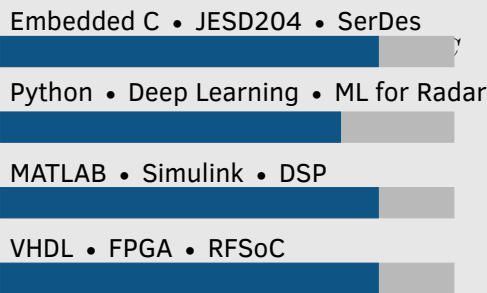
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Technical Skills

Overview



Programming



Education

M.Tech., Communication & Signal Processing
Indian Institute of Technology (IIT) Madras
CGPA: 9.32 / 10

B.Tech., Electronics & Communication Engineering
Bundelkhand Institute of Engineering & Technology
80% Graduated with Honours

About Me

Principal FPGA and AESA Radar Systems Engineer with **11 years of experience** in the architecture, design, integration, and deployment of mission-critical defence radar systems. Expertise spans digital system architecture, FPGA development, RF and microwave engineering, mixed-signal hardware, and real-time radar signal processing. Proven track record of delivering complex defence programs by translating operational requirements into scalable, high-performance hardware and FPGA solutions while ensuring system reliability, maintainability, and production readiness.

Led the development of large-scale **160 × 16 coherent-channel AESA radar systems** using direct RF sampling architectures, deterministic JESD204B/C synchronization, adaptive digital beamforming, and high-speed FPGA processing. Proven ability to translate complex system requirements into scalable hardware and FPGA architectures while leading multidisciplinary engineering teams from concept through production, system integration, customer acceptance, qualification testing, and field deployment.

Specialized in FPGA architecture, RFSoc platforms, radar waveform generation, high-speed digital interfaces, RF front-end design, and AI/ML-based radar target detection and classification. Experienced in technical leadership, architecture reviews, customer engagement, engineering mentoring, risk management, technical decision-making, and cross-functional program execution. Passionate about developing next-generation radar technologies through innovation, system optimization, and continuous improvement across the complete product development lifecycle.

Experience -- 11 Years

- Nov 2023 – Present
- Deputy Manager – Radar Systems R&D** Bharat Electronics Limited (BEL)
- Technical Lead for the architecture, development, integration and validation of large-scale AESA radar systems comprising **160 × 16 coherent radar channels**.
 - Defined system architecture for digital, RF, FPGA and mixed-signal subsystems supporting mission-critical surveillance radar applications.
 - Architected deterministic **multi-board synchronization** using JESD204B/C, enabling coherent operation across distributed radar processing units.
 - Designed direct RF sampling architectures integrating RF ADCs and DACs operating up to **12 GSps**.
 - Developed FPGA-based real-time radar processing including Digital Down Conversion (DDC), Pulse Compression, FFT, Doppler Processing, CFAR, Adaptive Digital Beamforming (PB & PA), and agile waveform generation.
 - Designed scalable high-speed data acquisition and deterministic streaming architectures using DDR4, 10 Gigabit Ethernet, AXI4-Stream and high-speed serial interfaces.
 - Achieved receiver sensitivity of **-154 dBm** through optimized RF architecture, ultra-low phase noise clocking and digital processing.
 - Led FPGA implementation including RTL development, IP integration, timing closure, CDC analysis, floorplanning, hardware validation and production release.
 - Applied AI/ML techniques for radar target detection, classification and clutter suppression.
 - Prepared system requirements, interface control documents (ICDs), design specifications, verification plans, and customer technical documentation.
 - Led architecture reviews, customer demonstrations, factory acceptance testing, field validation and mentored multidisciplinary engineering teams.
 - Coordinated cross-functional teams including hardware, firmware, FPGA, RF, mechanical, and manufacturing engineers.

Cross-Functional Collaboration

- RF & Microwave Engineering
- FPGA & Digital Design
- Embedded Firmware Development
- Software Engineering
- Signal Processing Algorithm Team
- Data Acquisition Team
- PCB Layout / CAD Design Team
- Mechanical Engineering
- Thermal Engineering
- Manufacturing Engineering
- Production & Operations
- Quality Assurance
- System Engineering
- Integration & Verification

Core Competencies

System Architecture

- Digital System Architecture
- AESA Radar System Architecture
- System Engineering
- Requirements Analysis
- Interface Definition (ICDs)
- Technical Roadmaps
- Product Development Lifecycle

FPGA & Digital Design

- VHDL, Verilog & RTL Design
- FPGA Architecture & Microarchitecture
- IP Integration
- Timing Closure & CDC/RDC Analysis
- Floorplanning & Resource Optimization
- FPGA Verification & Hardware Debugging

FPGA Platforms & Development Tools

- AMD/Xilinx RFSoc
- AMD/Xilinx Kintex UltraScale
- Intel (Altera) FPGA
- Vivado, Vitis & IP Integrator
- ModelSim / QuestaSim
- Tcl Scripting

2017 –

2022

Senior Hardware & Digital Systems Engineer Bharat Electronics Limited (BEL)

- Designed FPGA-based digital architectures for radar receiver, exciter and digital beamforming subsystems.
- Developed coherent radar signal processing algorithms including Pulse Compression, MTI, FFT, Doppler Processing and CFAR.
- Performed radar performance analysis including radar equation, SNR, receiver sensitivity, dynamic range and phase noise optimization.
- Executed system integration, hardware bring-up, laboratory validation, environmental qualification and field trials.
- Collaborated with RF, FPGA, software and manufacturing teams throughout product development and production.
- Supported customer demonstrations, technical reviews and product qualification activities.

2015 –

2016

Mixed Signal Processing & RF Design Bharat Electronics Limited (BEL)

- Designed mixed-signal hardware for RF transceiver systems.
- Designed RF exciter and receiver subsystems, frequency synthesis architectures, up/down converters, LO distribution, and ultra-low phase noise clocking systems.
- Performed schematic design, component selection, PCB reviews, and hardware validation.
- Conducted receiver chain analysis including gain, noise figure, phase noise, and sensitivity calculations.
- Supported SI/PI analysis, laboratory testing, and manufacturing support.

Research

2022 – 2024 **Postgraduate Research – Radar AI** Indian Institute of Technology Madras

Thesis: Target Detection and Classification Using Deep Learning for Surveillance Radar

- Deep learning models for radar target detection and classification
- Analysis of phase noise and micro-Doppler effects on ML-based radar systems
- Tools: MATLAB, Python, Deep Learning Toolboxes

Publications

- **Nirbhay Kumar Singh**, “Advanced Deep Learning Technique for Digital Predistortion to Improve Power Amplifier Linearity”. IEEE MAPCON2024
- **Nirbhay Kumar Singh**, Malwinder Singh, Shaik Abdul Subhan, Seema, “Phase Noise Effects on Micro-Doppler Extraction for Radar Target Classification Using Machine Learning”. IEEE APMC2024, Bali, Indonesia
- **Nirbhay Kumar Singh**, “Multi-board Synchronization of n-channel Direct RF Digital Transceiver for Active Phased Array Radar using JESD204 Interface”. IEEE I2CT · Apr 6, 2024
- **Nirbhay Kumar Singh**, Vikas Kumar, T Venakatamuni, “Ultra-Low Phase Noise, Digital Transceiver for High Range Resolution Pulsed Agile Radar”. 8th IEEE International Radio Conference, Mauritius · May 1, 2023
- **Nirbhay Kumar Singh**, T Venakatamuni, “A Wide-band SWaP System for VSWR and Power Measurement with Temperature Compensation Technique”. IEEE MAPCON2022 · Dec 14, 2022
- **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakatamuni, A Programmable, Multimode Operational 3U-VPX Based Digital Transceiver & Processing Module For CIT-MKXIIA IFF
- **Nirbhay Kumar Singh**, T Venakatamuni, “A method to Enhance Power Amplifier Linearity using Time delay estimation & correction Technique”, IEEE, 2022 Inter-

Radar Signal Processing

- Digital Down Conversion (DDC)
- Digital Pulse Compression (DPC)
- FFT & Doppler Processing
- MTI / MTD
- CFAR Target Detection
- Adaptive Digital Beamforming
- Pulse, FMCW, Chirp & Agile Waveforms
- Target Detection & Tracking

RF & Microwave Engineering

- Radar Link Budget Analysis
- Receiver Sensitivity (MDS)
- Gain & Noise Figure Analysis
- Phase Noise Optimization
- RF Exciter & Receiver Design
- Frequency Synthesis
- Up/Down Conversion
- RF ADC/DAC Integration
- Direct RF Sampling

Mixed-Signal Hardware

- Schematic Design
- PCB Design Review
- Component Selection
- Signal Integrity (SI)
- Power Integrity (PI)
- Clock Tree Design
- Hardware Bring-up & Validation

High-Speed Interfaces

- JESD204B/C
- DDR3/DDR4
- PCI Express
- Gigabit & 10 Gigabit Ethernet
- AXI4 / AXI4-Stream
- SPI, I²C, UART, GPIO

Artificial Intelligence

- Machine Learning for Radar
- Deep Learning
- Radar Target Detection
- Radar Target Classification
- Clutter Suppression

national Conference for Advancement in Technology (ICONAT) Goa, India. Jan 21-22, 2022

- Vikas Kumar, **Nirbhay Kumar Singh**, T Venakataamuni, "HIGH-RESOLUTION FMCW GENERATION FOR MICRO DOPPLER IN DRONE DETECTION RADAR", International Radar Symposium India(IRSI) 2019 **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakataamuni, "Performance Analysis of High-Resolution Radar Waveforms for Naval Tracking Radar". EWCI 2018
- **Nirbhay Kumar Singh**, Fouziya C, Venkataamuni, "Radar Target Emulator For Naval Tracking Radar Using FPGA", URSI-RCRS 2017

Patents

- A method and system of synchronization of an n-channel direct RF digital transceiver with high-speed serial JESD204 protocol. Patent no: 202341083757
- A Method to generate a wideband exciter signal having multi-pulse-coded waveforms with test-target simulation in a radar receiver. Patent no: 202341004980
- Wide-band SWAP System and Method for VSWR and Power Measurement using Digital Signal Processing with Temperature Compensation. Patent no: 202241070066
- Compact and Programmable Digital Transceiver for Software-defined Radar. Patent no: 202041013933
- System and Method to Convert Non-Coherent FMCW Radar to Coherent Radar. Patent no: 202141009908
- A SYSTEM AND METHOD FOR REALIZING ULTRA-LOW PHASE NOISE, FAST SETTLING, COMPACT PHASE COHERENT AND ENERGY EFFICIENT DIGITAL SOURCE GENERATOR. Patent no: 202341030479

Awards

- Nominated for the **IETE Young Scientist Award**, one of India's highest honors recognizing outstanding contributions by young scientists..
- R&D Excellence award 2020-23(Excellent Technology project): MKXIIA-CIT.
- R&D Excellence award 2020-23(Innovation category): RFSoc based Integrated Processor Module.
- R&D Excellence award 2019-20(Innovation category): Programmable RF and SoC based processing module for IFF MKXIIA - CIT.
- R&D Excellence award 2016-17(Innovation category): DIRECT SAMPLING DIGITAL RECEIVER for IFF.
- Best Technical Poster Awards (Defence AI, Unmanned Systems)

Additional Information

- Languages: English (Professional), Hindi (Native)
- Open for relocation.
- 11 years of experience in Defence & Aerospace R&D